

Chapter 1 Structure of Ceramic Materials

1.1 Structure of Crystals

1.2 Structure of Glass

1.3 Structural Imperfections

1.4 Surfaces, Interfaces, and Grain Boundaries

1.5 Atom Mobility (Diffusion)

Atomic Structure

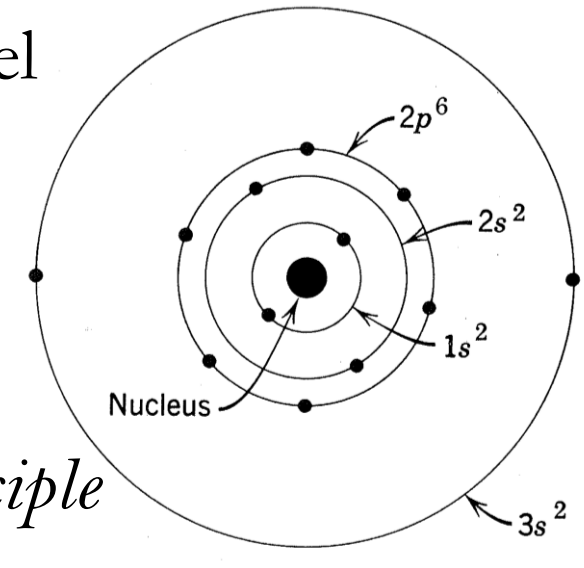
📌 Bohr Atomic Model

📌 Electron Orbits

Pauli exclusion Principle

📌 Electron Waves

📌 Quantum Numbers

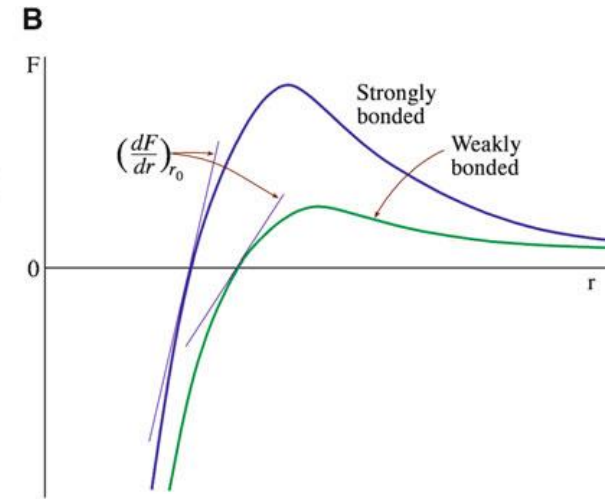
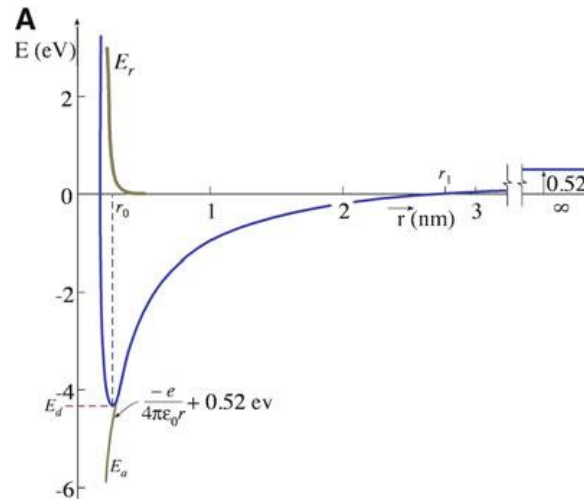
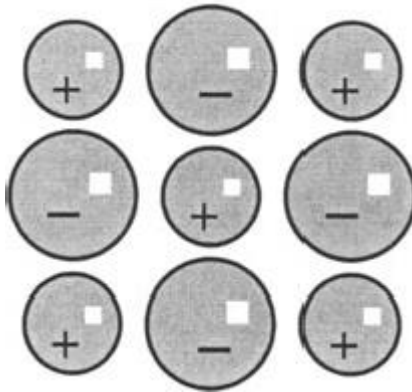


	$l=0$	1	2	3	4
$n=1$	1s	1p	1d	1f	1g
2	2s	2p	2d	2f	2g
3	3s	3p	3d	3f	3g
4	4s	4p	4d	4f	4g
5	5s	5p	5d	5f	5g
6	6s	6p	6d	6f	6g
7	7s	7p	7d	7f	
8	8s	8p	8d		

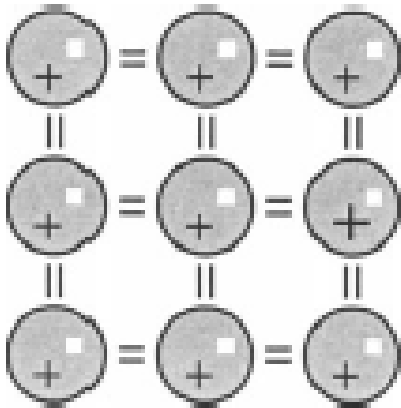
Interatomic Bonding

Primary Bonding

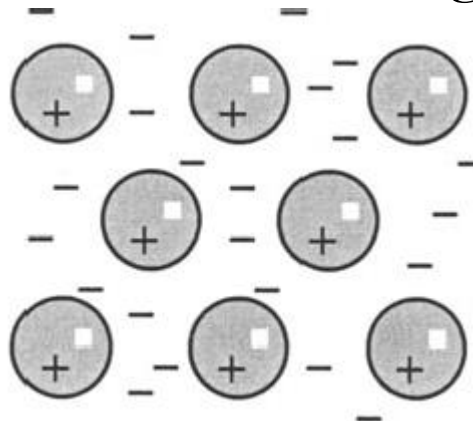
📌 Ionic Bonding



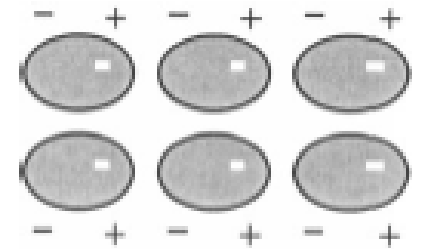
📌 Covalent Bonding



📌 Metallic Bonding



Secondary Bonding



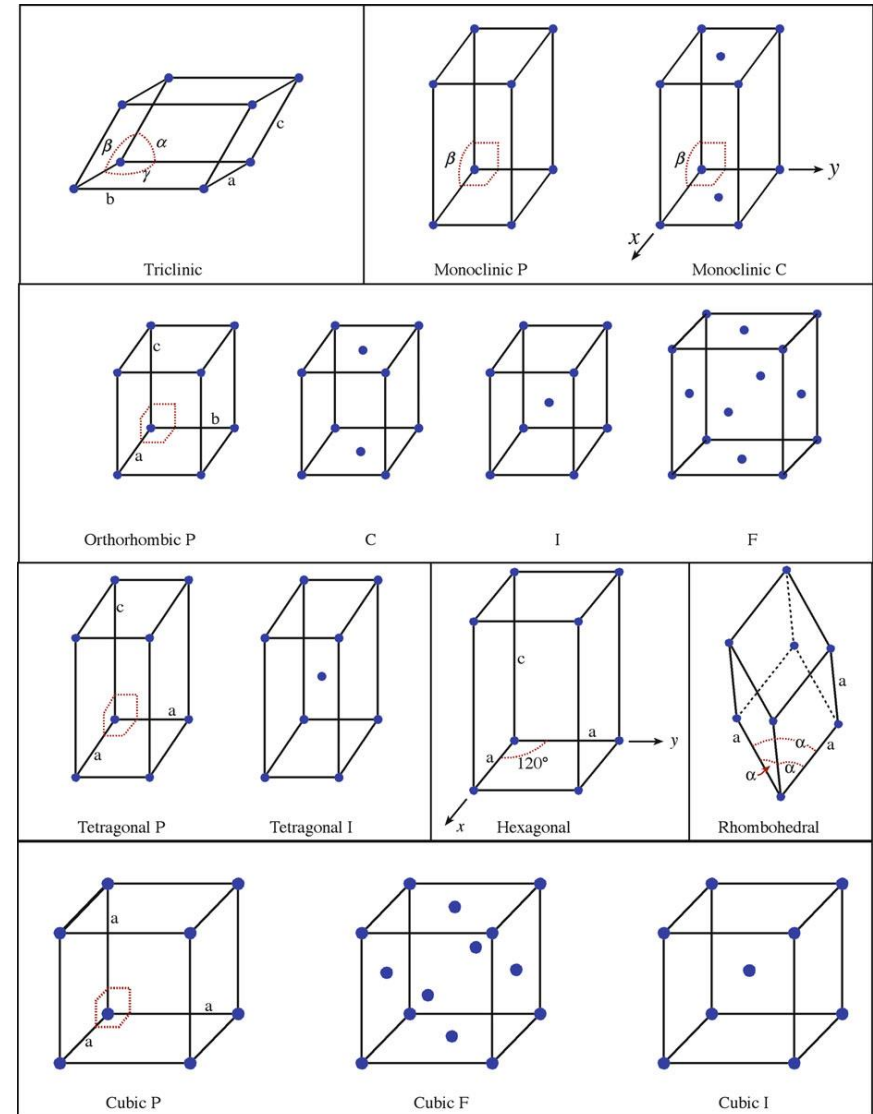
d van der Waals bonding

Crystal Structure

The 7 Crystal Systems

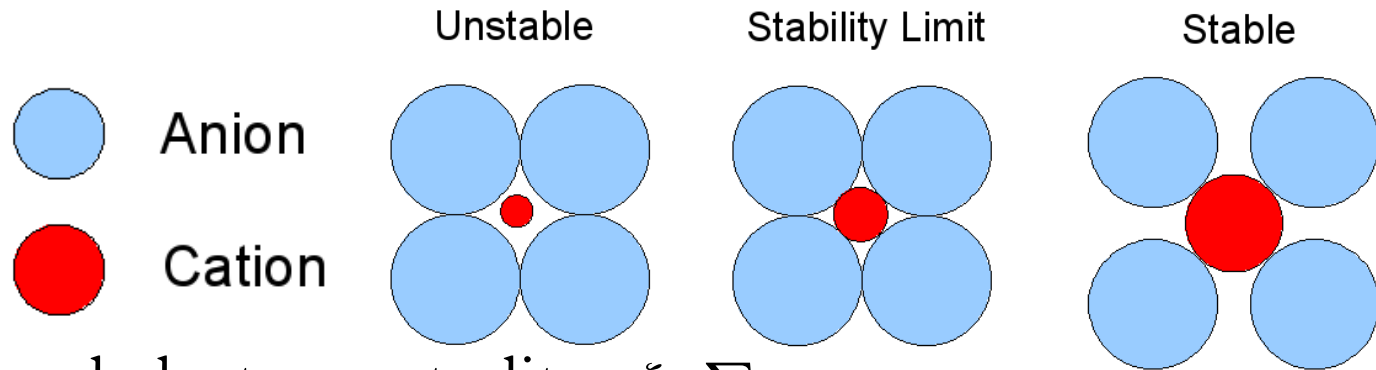
System	Relationship between lattice parameters	Example
Triclinic	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	Turquoise Kyanite Albite feldspar
Monoclinic	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ; \beta \neq 90^\circ$	Monazite Titanite Orthoclase
Orthorhombic	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	Olivine Brookite Stibnite
Tetragonal	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	Zircon Rutile Hausmannite
Hexagonal	$a = b \neq c$ $\alpha = \beta = 90^\circ; \gamma = 120^\circ$	High quartz Wurtzite Beryl
Rhombohedral (or trigonal)	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	Ilmenite Calcite Corundum
Cubic	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Halite Magnetite Garnet

The 14 Bravais Lattices



Pauling's Rules

Rule 1: Each cation is surrounded by coordinated anions which form a polyhedron. $r_c + r_a \rightarrow d_{a-c}$; $r_c / r_a \rightarrow C.N.$



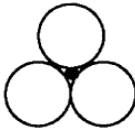
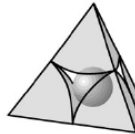
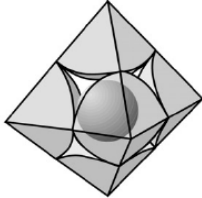
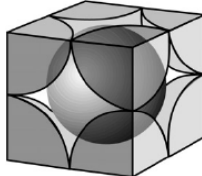
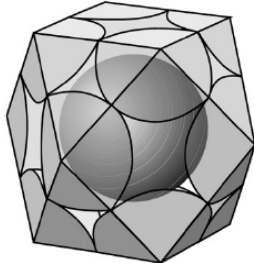
Rule 2: Local electroneutrality $\xi = \sum_i s_i$

Rule 3: The polyhedra in a structure tend not to share edges or faces.

Rule 4: The cation of high valency and small C.N. tend not to share polyhedron elements with one another.

Rule 5: The number of essentially different kinds of constituents in a crystal tends to be small

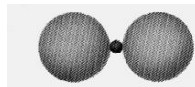
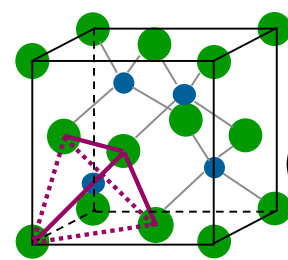
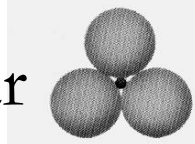
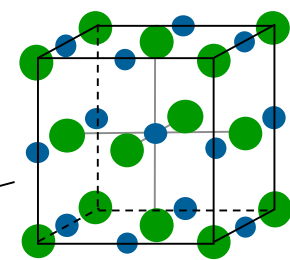
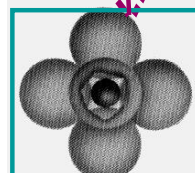
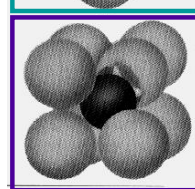
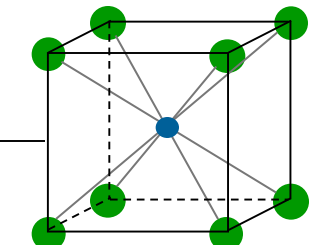
Pauling's Critical Radius Ratios

Coordination number	Arrangement of ions around central ion	Range of cation/anion ratios	Structure
3	corners of a triangle	≥ 0.155	
4	corners of a tetrahedron	≥ 0.225	
6	corners of a octahedron	≥ 0.414	
8	corners of a cube	≥ 0.732	
12	corners of a cuboctahedron	≈ 1.000	

Site Selection

- C. N. increases as r_c/r_a increases

Issue: How many anions can you arrange around a cation?

r_c/r_a	C.N.				
< 0.155	2	Linear			ZnS (zinc blende)
$0.155 - 0.225$	3	Triangular			
$0.225 - 0.414$	4	T_D			NaCl
$0.414 - 0.732$	6	O_H			
$0.732 - 1.0$	8	Cubic			CsCl

Example: Predicting Structure of FeO

- On the basis of ionic radii, what crystal structure would you predict for FeO ?

Cation Ionic radius (nm)

Al³⁺ 0.053

Fe²⁺ 0.077

Fe³⁺ 0.069

Ca²⁺ 0.100

Anion

O²⁻ 0.140

Cl⁻ 0.181

F⁻ 0.133

- Answer:

$$\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.077}{0.140} = 0.55$$

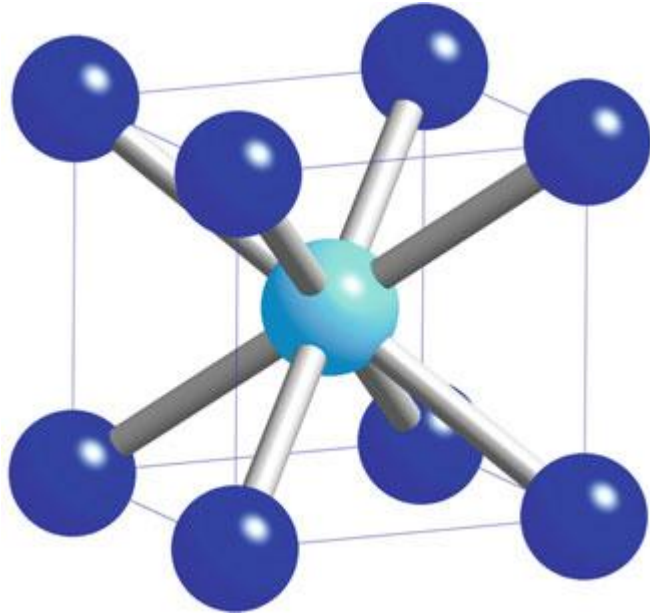
based on this ratio,

--C.N. = 6

--Structure = NaCl

Typical Crystal Structures of Ceramic Materials-MX

CsCl

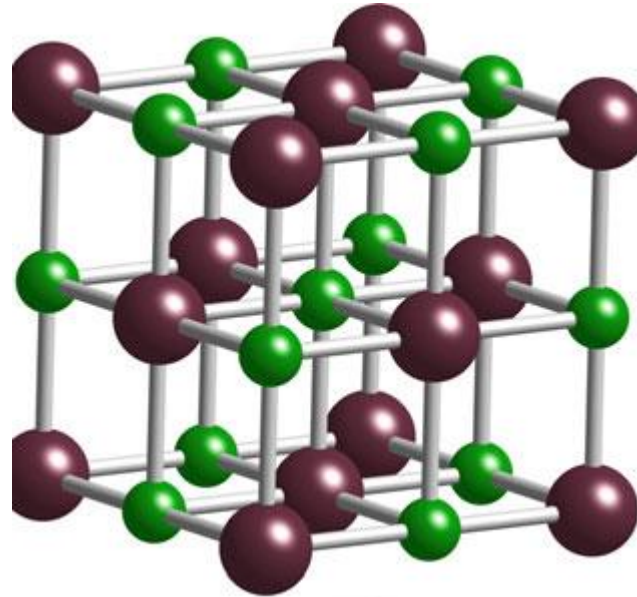


$$r_{\text{Cs}^+} / r_{\text{Cl}^-} = 171/181 = 0.94 > 0.732$$

8-fold Coordinated

NiAl, CsBr, CsI, TlCl...

NaCl



Na^+ (0.5,0,0)

Cl^- (0,0,0)

C.N.=6

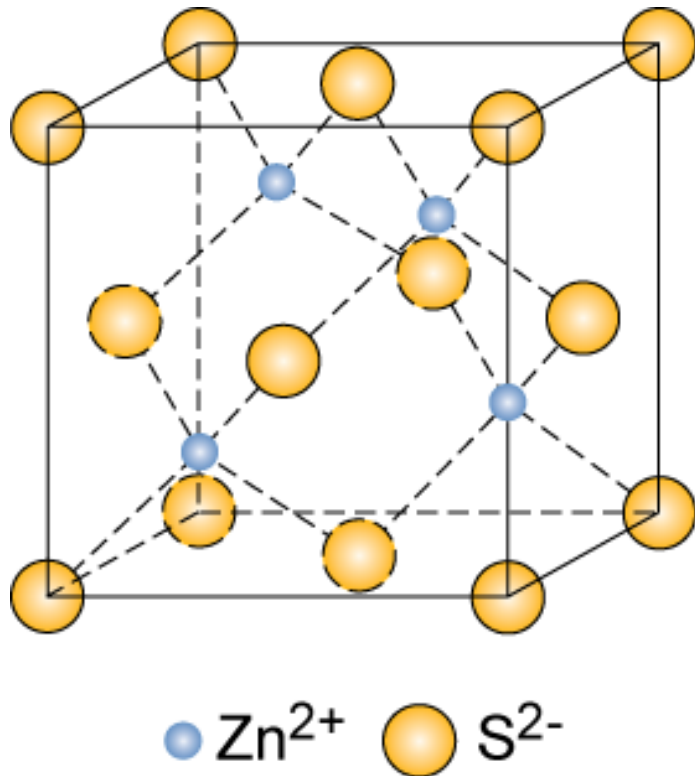
$$r_{\text{Na}^+} / r_{\text{Cl}^-} = 0.56 > 0.414 \text{ but } < 0.732$$

Sulfides, carbides,

some oxides, including MgO, CaO, SrO, BaO, CdO, FeO, and NiO.

Typical Crystal Structures of Ceramic Materials-MX

Zinc Blende Structure



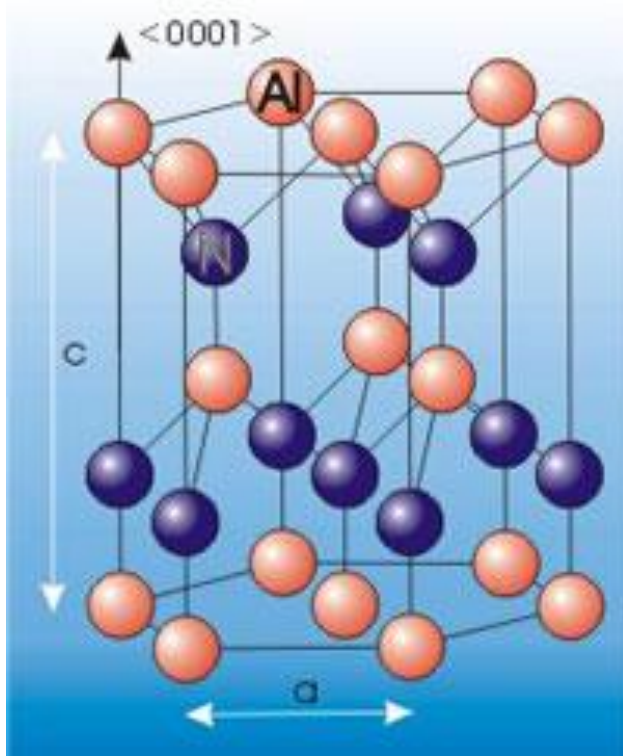
Here the small cations sit between 4 anions, $CN = 4$

$$0.225 < r_C / r_A < 0.414.$$

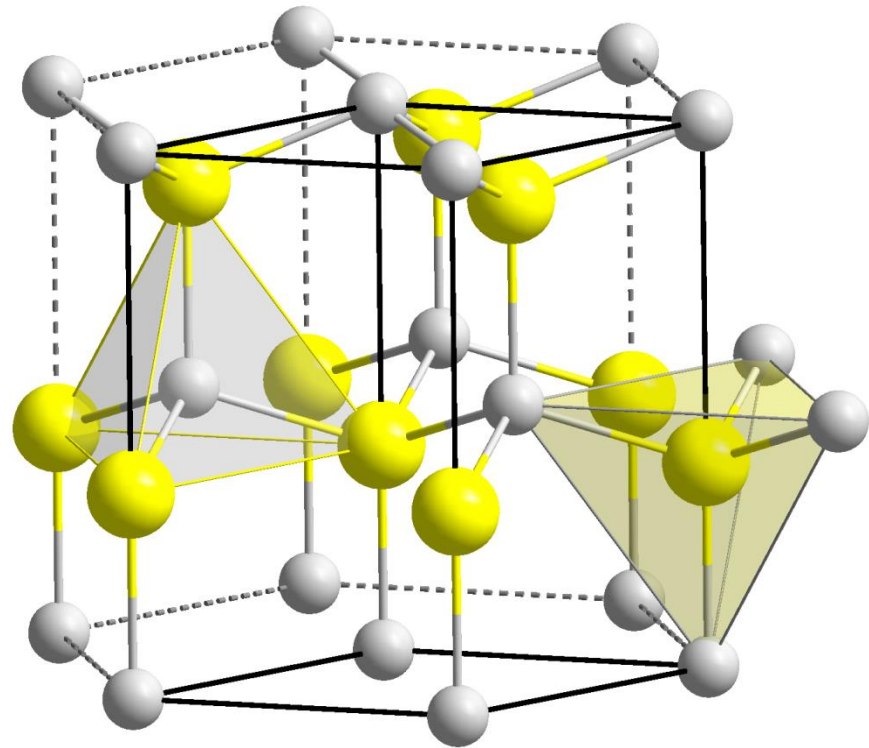
“Tetrahedral” interstitial site.
only 4 of the possible 8 tetrahedral holes is occupied.

Ex: ZnO, ZnS, SiC, GaAs, InP, InSb.....

Typical Crystal Structures of Ceramic Materials-MX



Aluminum nitride



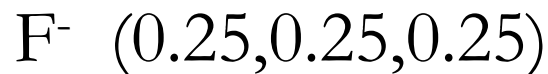
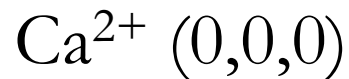
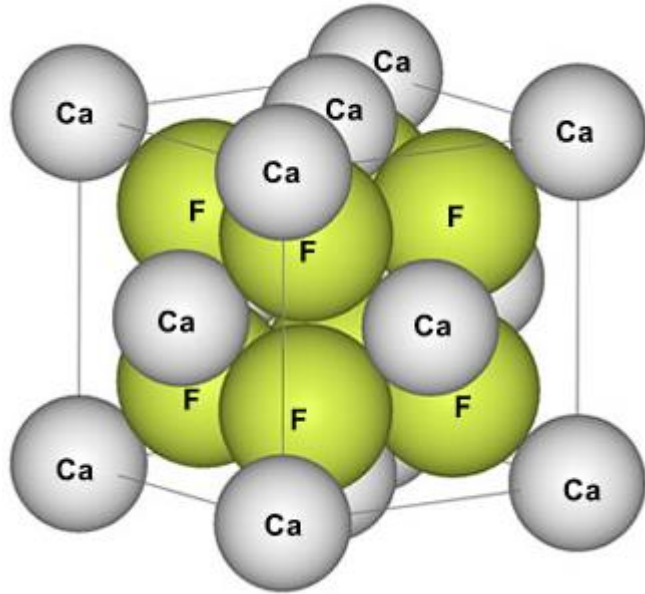
Wurtzite (Second polymorph of ZnS)

Wurtzite: ABABAB; Zinc blende: ABCABC

Ex: ZnS , AlN , BeO , GaN

Typical Crystal Structures of Ceramic Materials-MX₂

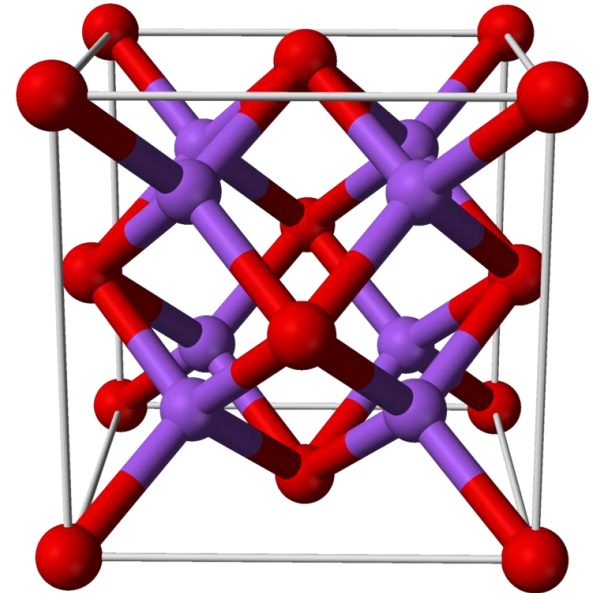
Fluorite (CaF₂)



$$r_{\text{Ca}^{2+}} / r_{\text{F}^-} = 100/130 = \square 0.8$$

Ex: c-ZrO₂; UO₂, CeO₂.....

Anti-Fluorite (M₂X)

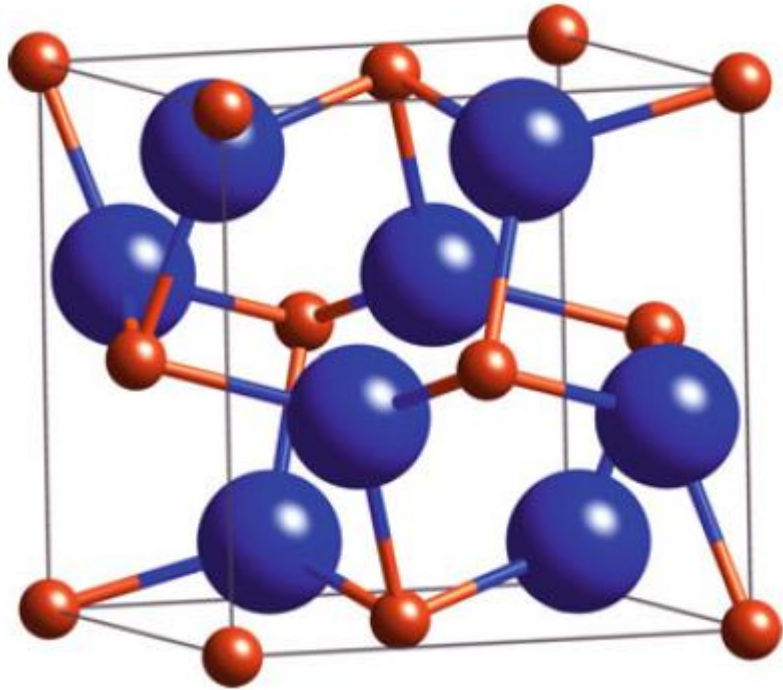


Reverse Cation-Anion



Ex: Li₂O, Na₂O, K₂O.....

Typical Crystal Structures of Ceramic Materials-MX₂



Pyrite (FeS₂)

What's the difference from NaCl ?

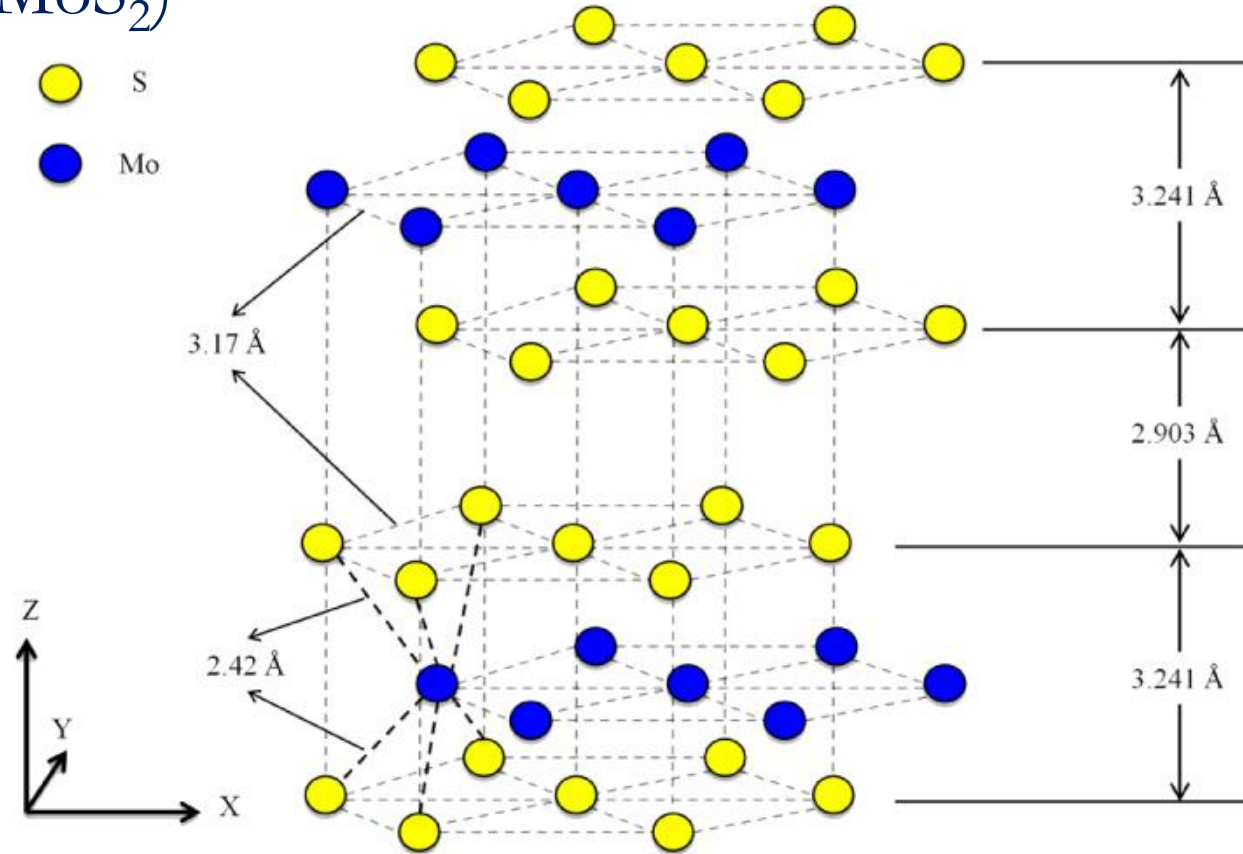
1. Both appear to have FCC, Cl $\rightarrow$ S₂ dumbbell; the dumbbells point along different directions for each of the edges; $m3m \rightarrow m3$

2. NaCl has a 4-fold axis along [001]

Ex: MX₂ of transition metals M and chalcogens (X = O, S, Se and Te.

Typical Crystal Structures of Ceramic Materials-MX₂

Molybdenite (MoS₂)



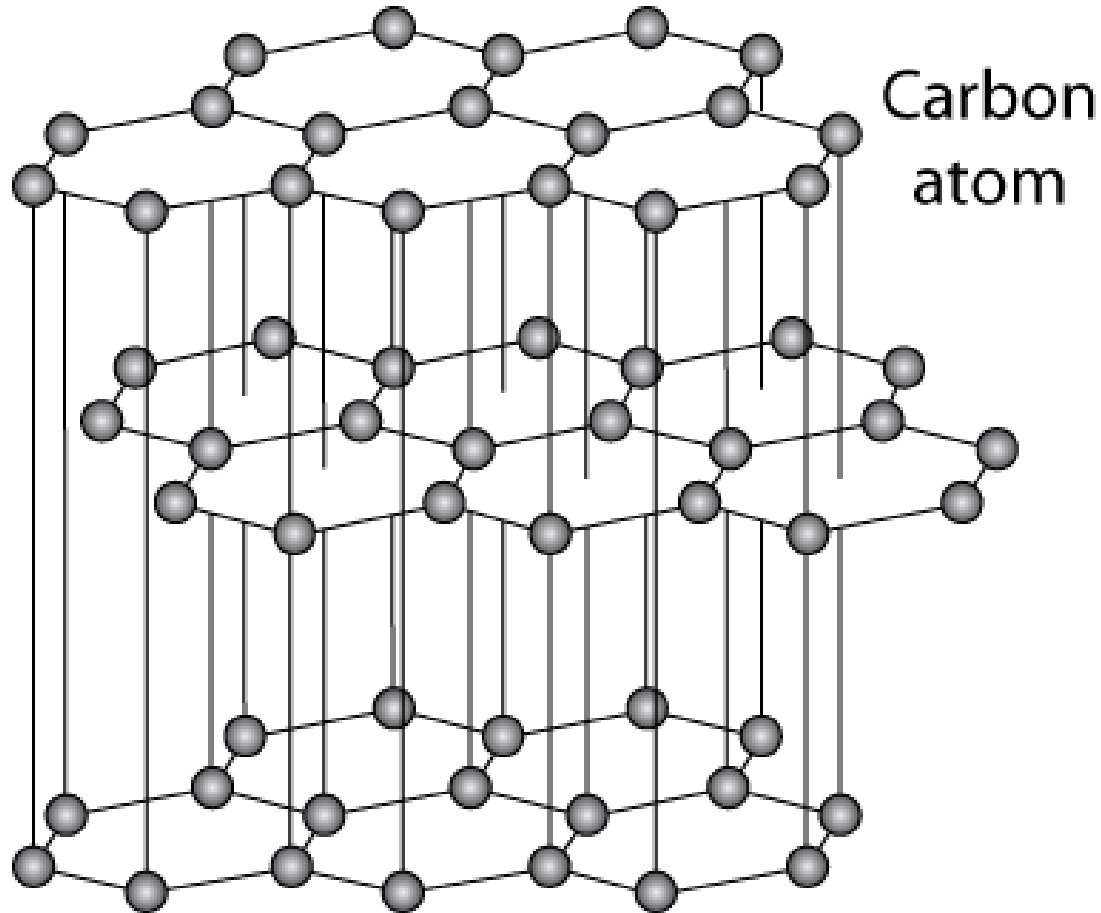
Stacking sequence: AbABaB (Capital: S; Lowercase : Mo)

Interlayer bonding: Van der Waals, very weak, solid lubricant

Ex: Mo and W chalcogenides

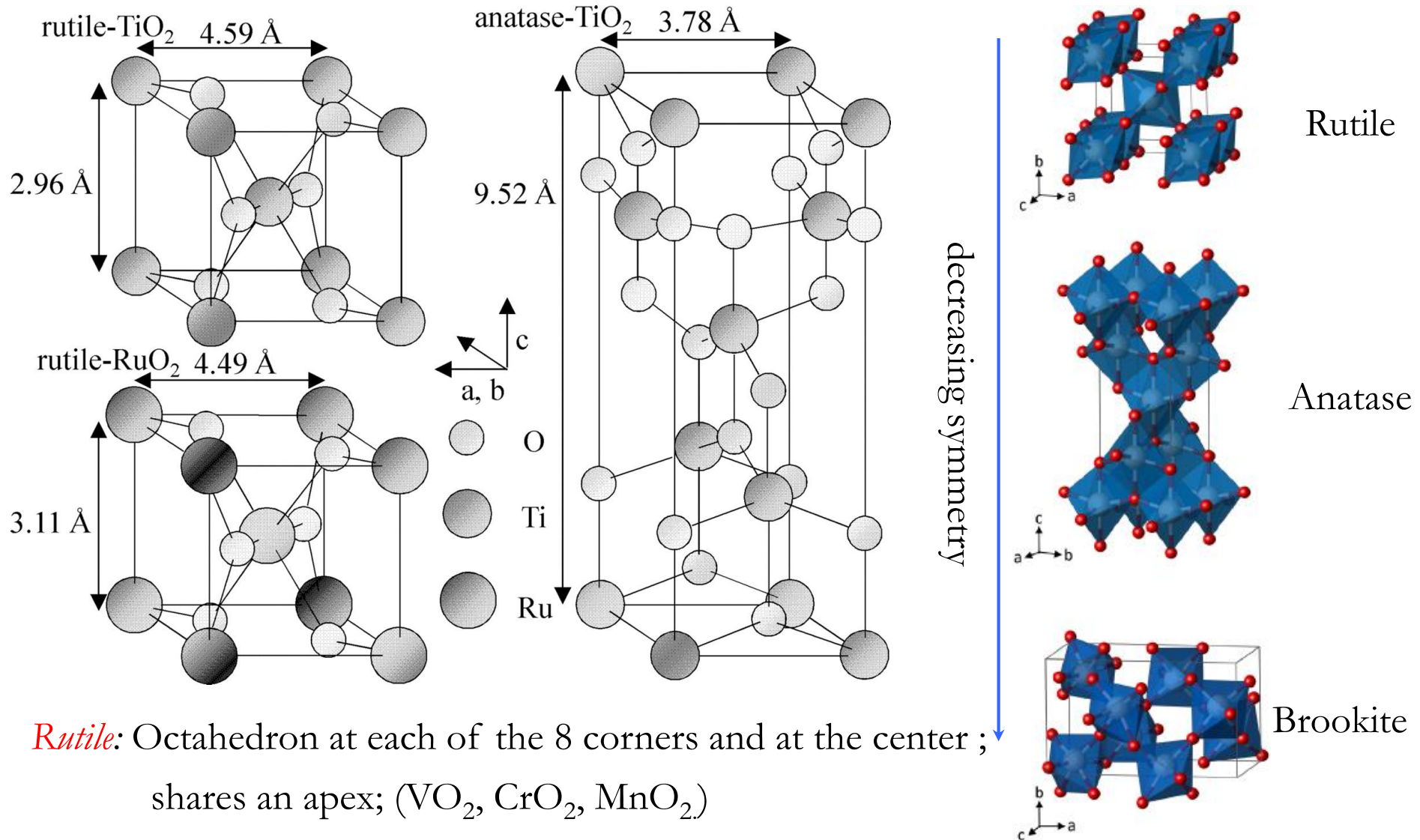
Similar Layered Structure-Graphite

- ✦ layer structure – aromatic layers



- 📄 Weak van der Waal's forces between layers
- 📄 Planes slide easily, good lubricant

Typical Crystal Structures of Ceramic Materials- AX_2

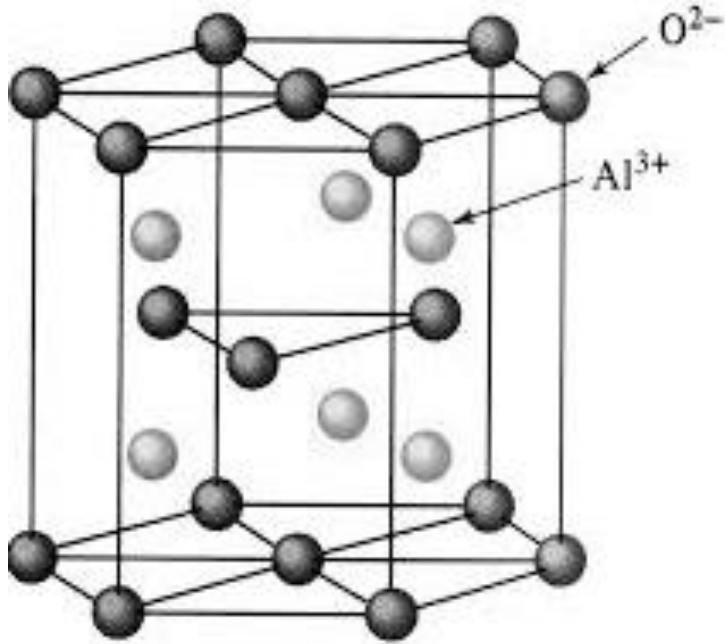


Rutile: Octahedron at each of the 8 corners and at the center ;
shares an apex; (VO_2 , CrO_2 , MnO_2)

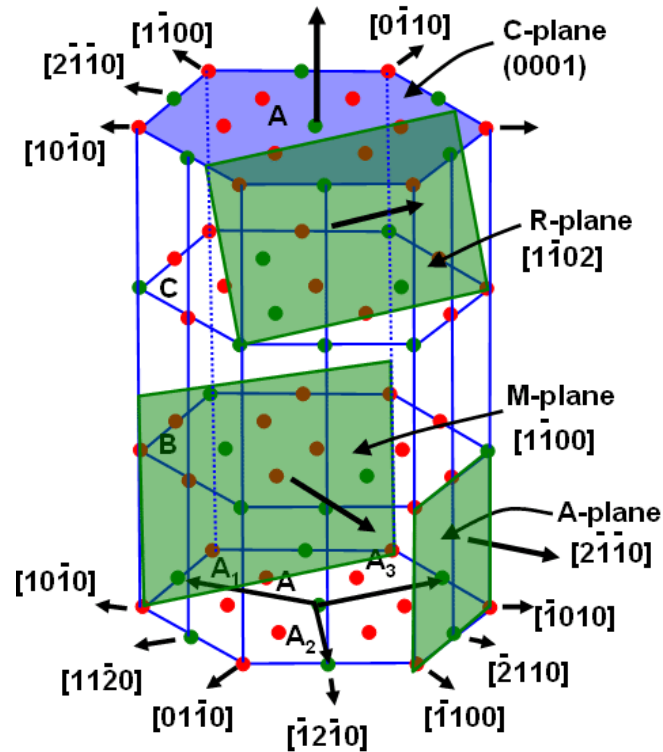
Anatase: Octahedron distorted and shares 4 of its edges with other octahedrons;

Brookite: shares both edges and corners

Typical Crystal Structures of Ceramic Materials- A_2X_3



Alumina (Al_2O_3)
Corundum structure

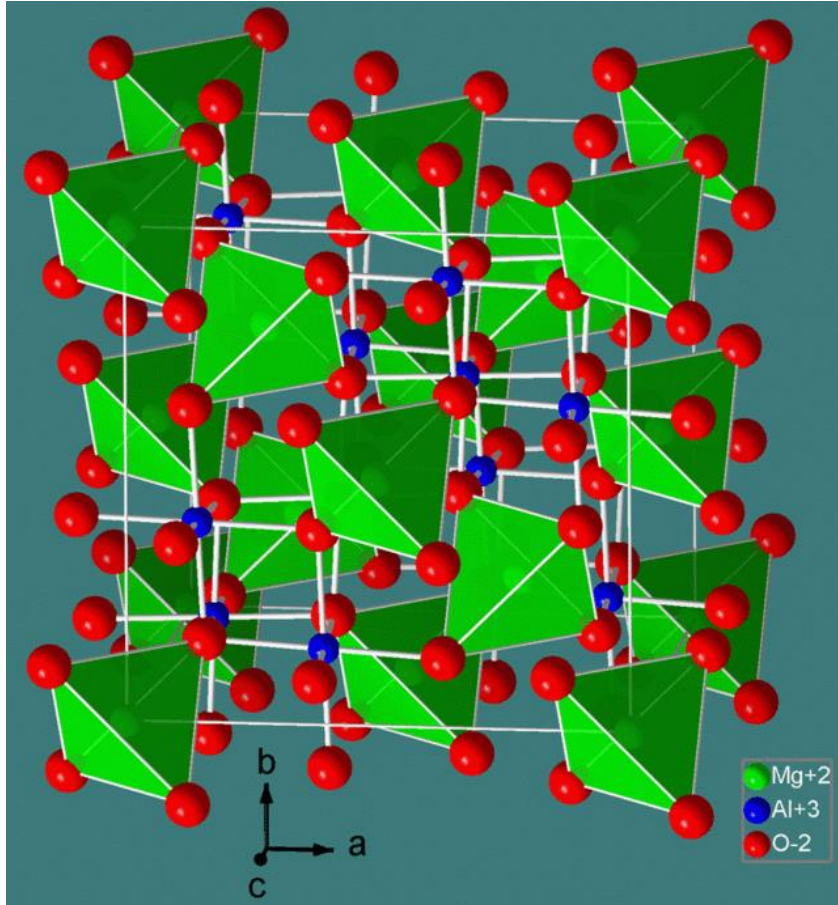


Sapphire

Alumina: trigonal, $\bar{3}m$, Pseudo-hexagonal, but not six-fold.

Typical Crystal Structures of Ceramic Materials- AB_2O_4

Spinel Structure ($MgAl_2O_4$)



1. Cubic crystal system,
2. the oxide anions in a cubic close-packed lattice;
3. the cations A and B occupying some or all of the octahedral and tetrahedral sites in the lattice.

Ex:

Al Spinel:

(Mg, Be, Zn, Fe, Mn, (Mg, Fe)) Al_2O_4

Fe Spinel:

(Cu, Mn, Ni, Fe, Ti, (Zn, Fe)) Fe_2O_4

Cr Spinel:

(Mg, Fe, Zn) Cr_2O_4

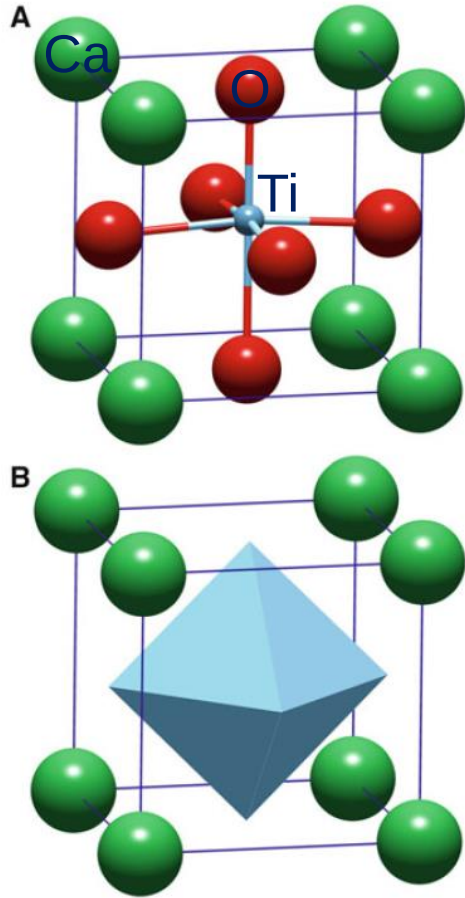
- *Normal spinel:* The A^{2+} ions occupy only tetrahedral sites, and the B^{3+} ions occupy only octahedral sites.
- *Inverse spinel:* All the A^{2+} ions and half the B^{3+} ions sit on the octahedral sites; the tetrahedral sites are occupied now by the other half of the B^{3+} ions.

O^{2-} : 140 pm, Mg^{2+} : 72 pm, Ni^{2+} : 69 pm, Co^{2+} : 74 pm

O^{2-} : 140 pm, Al^{3+} : 53 pm, Fe^{3+} : 65 pm, Cr^{3+} : 62 pm

Typical Crystal Structures of Ceramic Materials- ABO_3

Perovskite Structure (CaTiO_3)



BaTiO_3 AND KNbO_3

A	B	O
$R_{\text{Ba}^{2+}} = 136 \text{ pm}$	$R_{\text{Ti}^{4+}} = 61$	$R_{\text{O}^{2-}} = 140 \text{ pm}$
$R_{\text{K}^{+}} = 138 \text{ pm}$	$R_{\text{Nb}^{5+}} = 64$	$R_{\text{O}^{2-}} = 140 \text{ pm}$

The larger A^{2+} and the anion combine to form a close-packed arrangement, with smaller B_{4+} , sitting in the oxygen octahedral interstices

The bond strength is given as

$$\text{Ti} - \text{O} = +4/6 = 2/3; \quad \text{Ca} - \text{O} = +2/12 = 1/6$$

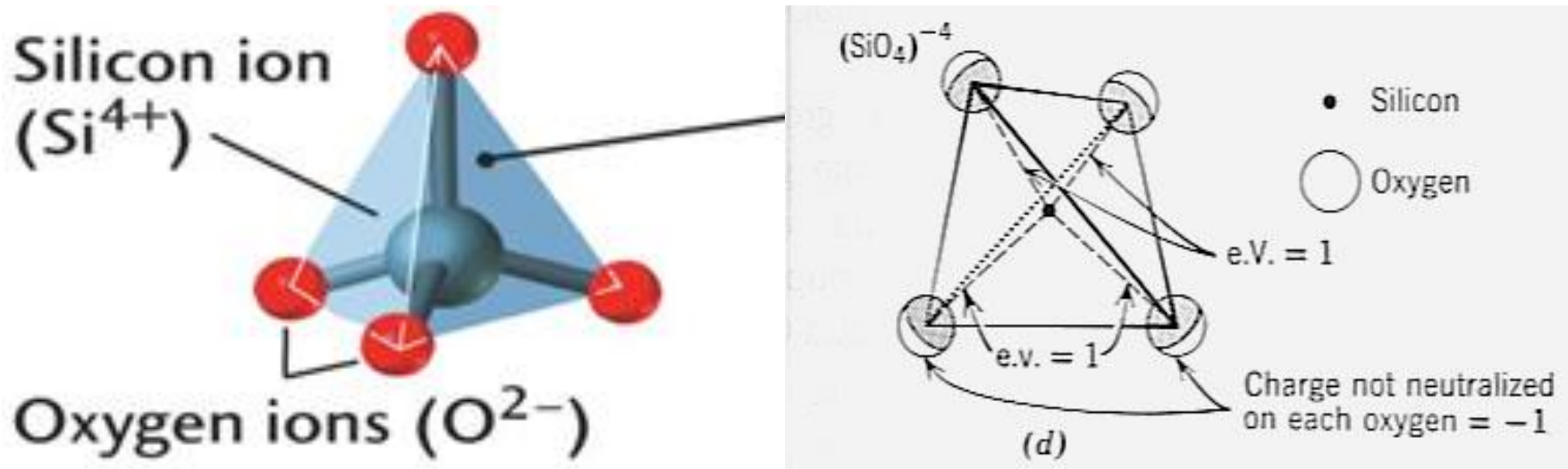
Each O^{2-} anion coordinates with two Ti^{4+} and four Ca^{2+} cations so that the total bond strength is

$$2 \times 2/3 + 4 \times 1/6 = +2$$

Ex: $(\text{Ba}, \text{Sr}, \text{Ca})\text{TiO}_3$, KNbO_3

The Silicates and Structures Based on SiO_4

Silicon Tetrahedra – the basic building block of silicate minerals

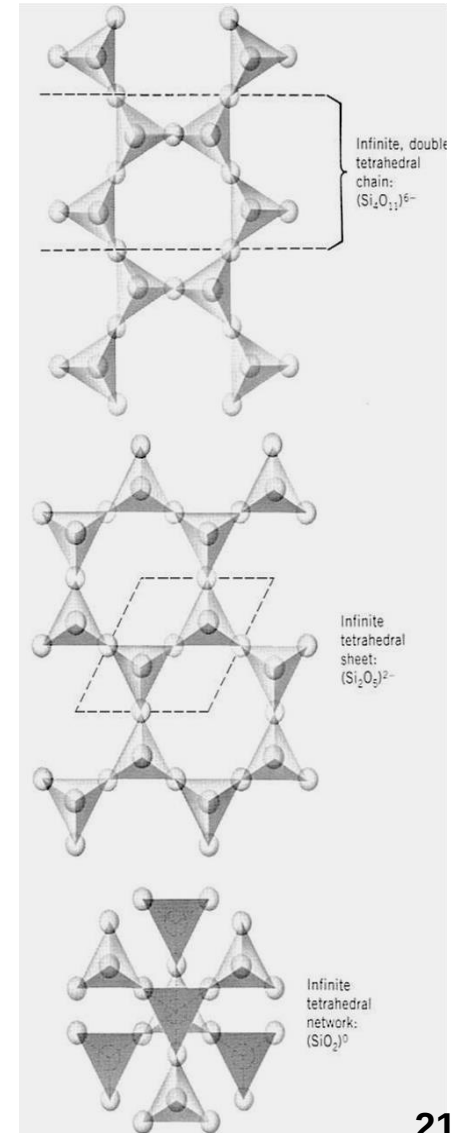
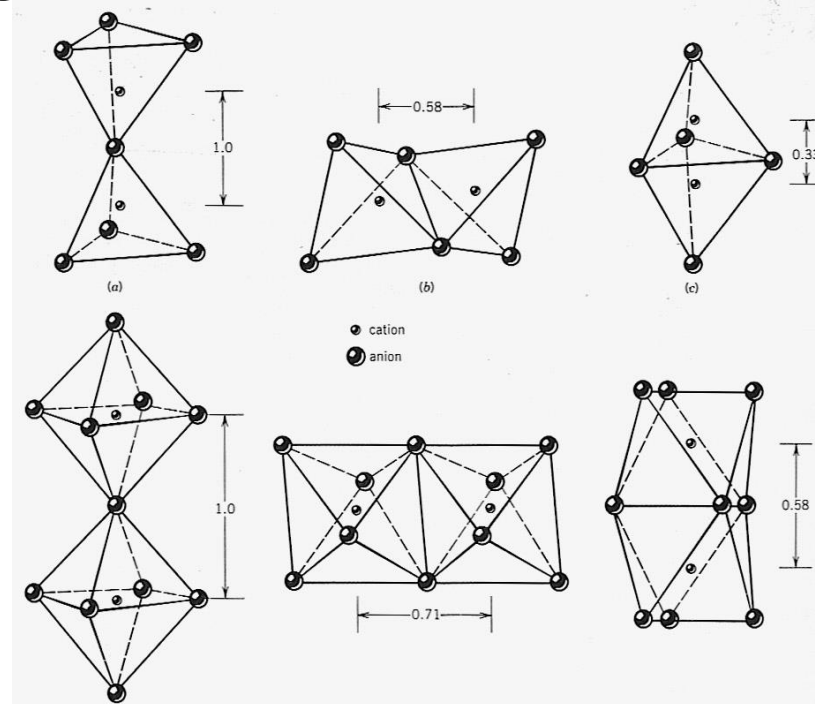
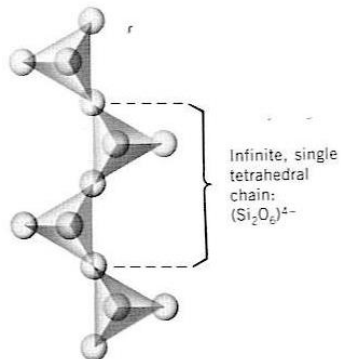
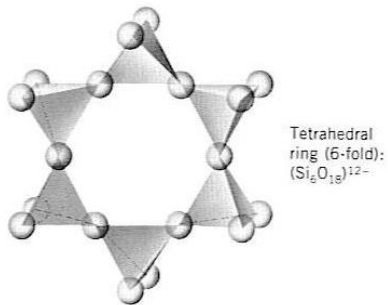
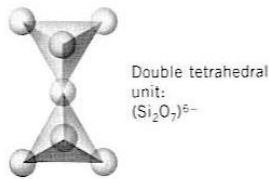
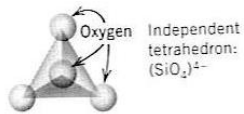


- The Si-O bond – 50% covalent, 50% ionic;
- Bond strength $= Z/\text{CN} = 4/4 = 1$;
- Each tetrahedral O shares a -1 charge with the tetrahedral Si and has an extra -1 charge to share with another cation;
- 4 oxygens in each tetrahedron, so total charge -4

Polymerization of Silicon Tetrahedra

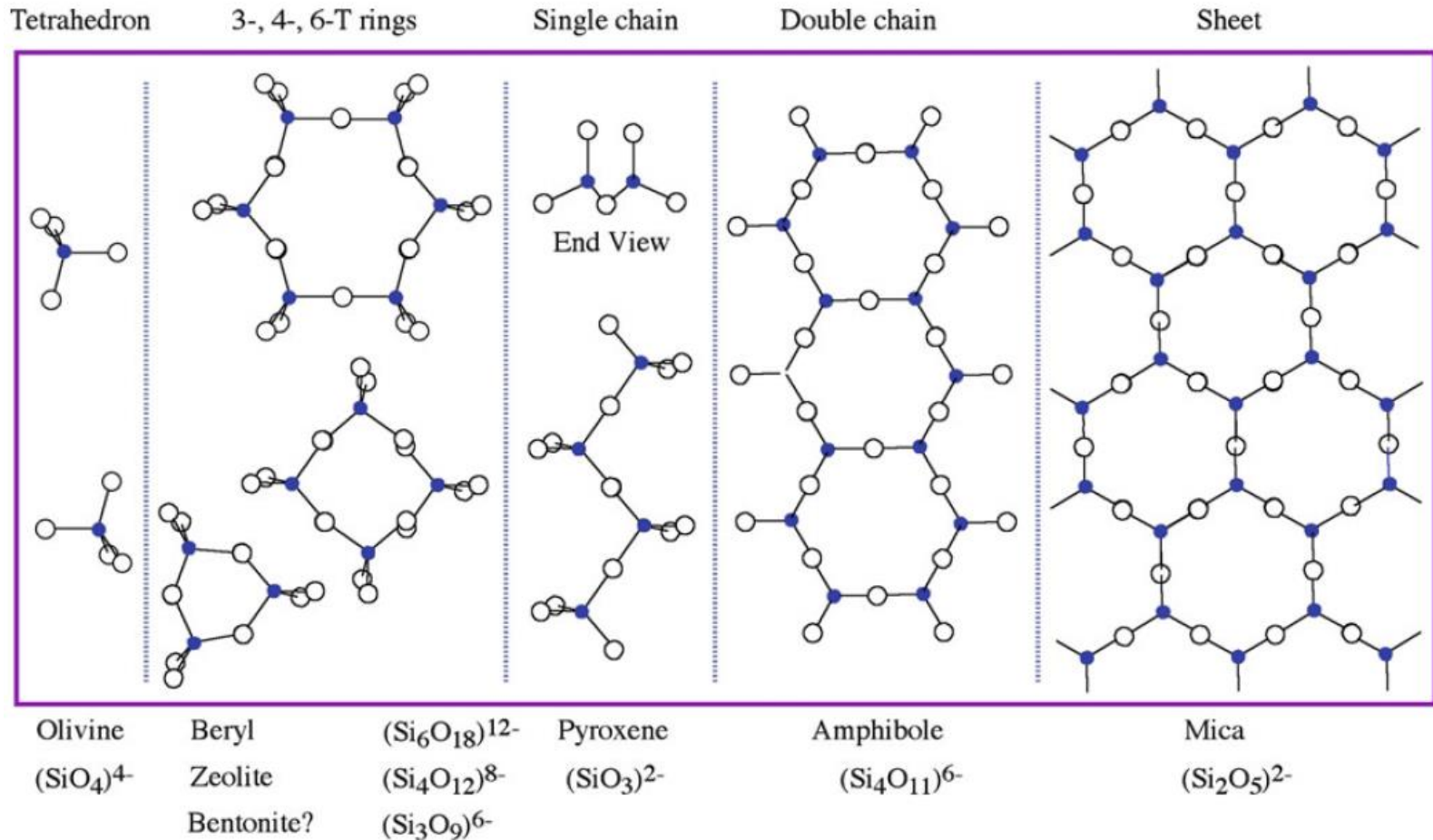
Oxygens can share electrons with two silicons

Adjacent Si tetrahedra can share corners, but because of the high repulsive charge of Si^{4+} cations, they will not share edges or faces. These shared corners are called bridging oxygens.



The Silicates and Structures Based on SiO_4

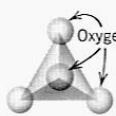
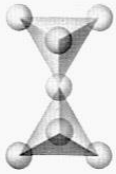
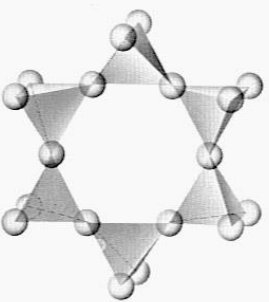
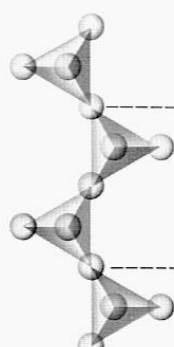
$r_{\text{Si}^{4+}}/r_{\text{O}^{2-}} = 0.40/1.40 = 0.29 < 0.41$ The SiO_4^{4-} : building blocks for all silicates

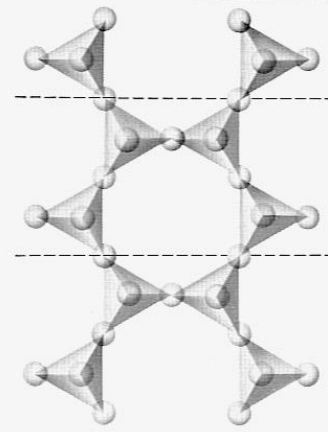
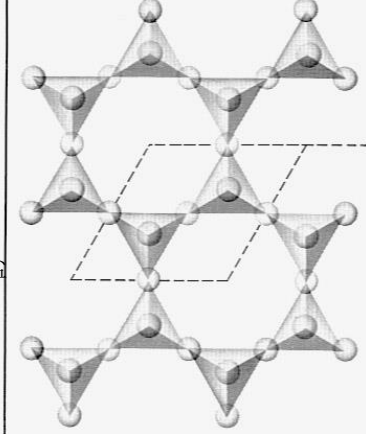
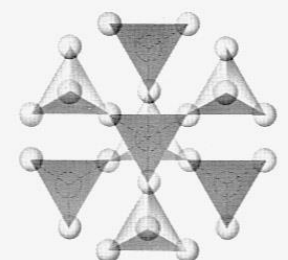


Silicate Mineral Classification

(based on arrangement of SiO_4 tetrahedra)

岛硅酸盐 (各 SiO_4 -4之间无共用氧原子)

Class	Arrangement of SiO_4 tetrahedra (central Si^{4+} not shown)	Unit composition	Mineral example
Nesosilicates		$(\text{SiO}_2)^{4-}$	Olivine, $(\text{Mg, Fe})_2\text{SiO}_4$ 镁质橄榄石
Sorosilicates 双硅酸盐		$(\text{Si}_2\text{O}_7)^{6-}$	Hemimorphite, $\text{Zn}_4\text{Si}_2\text{O}_7(\text{OH}) \cdot \text{H}_2\text{O}$ 异极矿
Cyclosilicates 环状硅酸盐		$(\text{Si}_6\text{O}_{18})^{12-}$	Beryl, $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ 绿柱石
Inosilicates (single chain) 链状硅酸盐		$(\text{Si}_2\text{O}_6)^{4-}$	Pyroxene e.g. Enstatite, MgSiO_3 辉石, 硬玉

Inosilicates (double chain)		$(\text{Si}_4\text{O}_{11})^{6-}$	Amphibole e.g. Anthophyllite, $\text{Mg}_7\text{Si}_8\text{O}_{22}(\text{OH})_2$ 闪石
Phyllosilicates 页硅酸盐		$(\text{Si}_2\text{O}_5)^{2-}$	Mica e.g. Phlogopite, $\text{KMg}_3(\text{AlSi}_3\text{O}_{10})(\text{OH})_2$ 云母
Tectosilicates 网硅酸盐		$(\text{SiO}_2)^0$	High cristobalite, SiO_2 方石英

Silicate Mineral Classification

(based on arrangement of SiO_4 tetrahedra)

Nesosilicates

Sorosilicates

Cyclosilicates



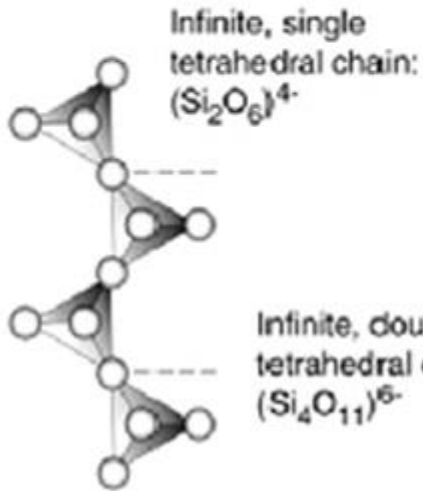
Isolated Tetrahedron
 $(\text{SiO}_4)^{4-}$



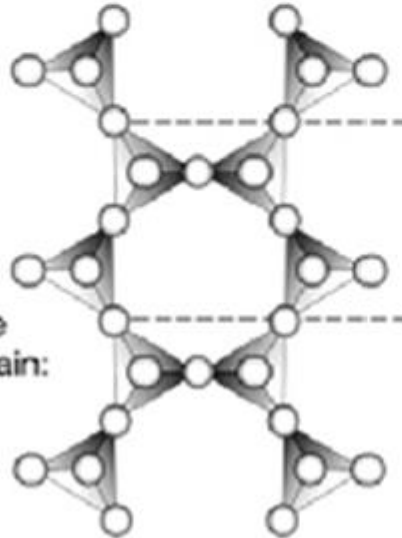
Double Tetrahedra
 $(\text{Si}_2\text{O}_7)^{6-}$



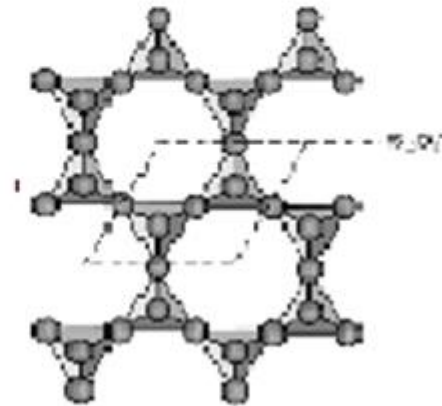
Tetrahedral
ring (6-fold):
 $(\text{Si}_6\text{O}_{18})^{12-}$



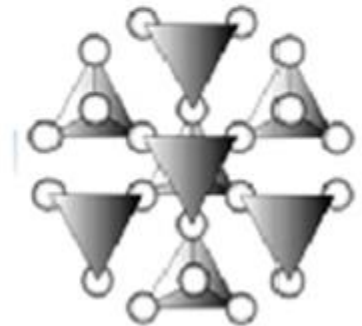
Infinite, single
tetrahedral chain:
 $(\text{Si}_2\text{O}_6)^{4-}$



Infinite, double
tetrahedral chain:
 $(\text{Si}_4\text{O}_{11})^{6-}$



Infinite tetrahedral sheets
 $(\text{Si}_2\text{O}_5)^{-2}$



Infinite tetrahedral
network:
 $(\text{SiO}_2)^0$

Inosilicates

Inosilicates

Phyllosilicates

Tectosilicates

Nesosilicates (independent tetrahedra)

- ✚ $X_2(\text{SiO}_4)$ Unit Composition X often +2 valence
- ✚ Isolated, but tightly packed $(\text{SiO}_4)^{4-}$ tetrahedra
- ✚ Forms silicate minerals with:
 - High density and hardness
 - Equi-dimensional habits
 - Poor cleavage
- ✚ Low degree of Al substitution with Si



☞ **Phenakite group** 硅铍石

Phenakite - Be_2SiO_4 ; Willemite- Zn_2SiO_4

☞ **Olivine group**

Forsterite- Mg_2SiO_4 ; Fayalite- Fe_2SiO_4 ; Tephroite- Mn_2SiO_4

☞ **Garnet group** 石榴石

Pyrope - $\text{Mg}_3\text{Al}_2(\text{SiO}_4)_3$; Almandine- $\text{Fe}_3\text{Al}_2(\text{SiO}_4)_3$;
Spessartine- $\text{Mn}_3\text{Al}_2(\text{SiO}_4)_3$; Grossular- $\text{Ca}_3\text{Al}_2(\text{SiO}_4)_3$...

☞ **Zircon group**: ZrSiO_4 锆石

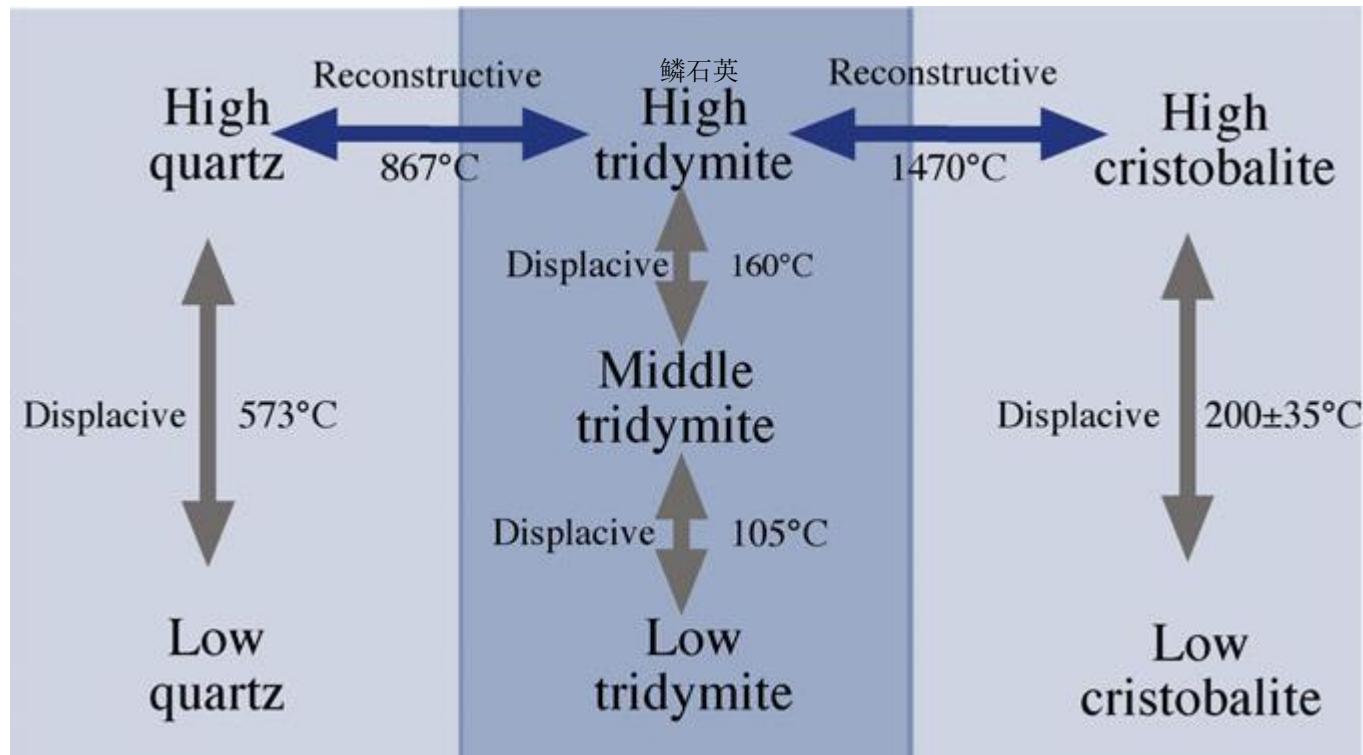
☞ **Al_2SiO_5 group**

Andalusite , Kyanite, Sillimanite

Typical Crystal Structures of Ceramic Materials-Polymorphs

✦ **Polymorphs:** Same chemical composition but different crystal structures.

Ex: Silica (SiO_2)



Silica conversion by displative/reconstructive structural transformations

Typical Crystal Structures of Ceramic Materials-Summary

Formula	Cation: anion coordination	Type and number of interstices occupied	Compact cubic stacking	Compact hexagonal stacking
MX	6:6	All the octa.	NaCl, FeO, MnS, TiC ZnS blende, CuCl, AgI- γ	NiAs, FeS, NiS
	4:4	Half the tetra.		ZnS würtzite, AgI- β
MX ₂	8:4	All the tetra.	CaF ₂ fluorine, ThO ₂ , ZrO ₂ , UO ₂	–
	6:3	Half the octa. Alternating layers with all the occupied sites	CdCl ₂	CdI ₂ , TiS ₂
MX ₃	6:2	1/3 of the octa. Alternating pairs of layers with 2/3 of the octa. occupied		BiI ₃ , FeCl ₃ , TiCl ₃ , VCl ₃
M ₂ X ₃	6:4	2/3 of the octa.		Al ₂ O ₃ corundum, Fe ₂ O ₃ , V ₂ O ₃ , Ti ₂ O ₃ , Cr ₂ O ₃
ABO ₃		2/3 of the octa.		FeTiO ₃ ilmenite
AB ₂ O ₄		1/8 of the tetra. and 1/2 of the octa.	MgAl ₂ O ₄ spinel, MgFe ₂ O ₄ inverse spinel	Mg ₂ SiO ₄ olivine